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Math 400

Section 2.2 The Limit of a Function

Objective 1: Introduction to Limits.

1 Intuitive Definition of a Limit Suppose $f(x)$ is defined when x is near the number a. (This means that f is defined on some open interval that contains a, except possibly at a itself.) Then we write

$$\lim_{x \to a} f(x) = L$$

and say "the limit of $f(x)$, as x approaches a, equals L"

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by restricting x to be sufficiently close to a (on either side of a) but not equal to a.

This says that the values of $f(x)$ approach L as x approaches a. In other words, the values of $f(x)$ tend to get closer and closer to the number L as x gets closer and closer to the number a (from either side of a) but $x \neq a$.

Example 1: Find the following limit, if it exists.

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$$

Objective 2: Left and Right Sided Limits.

2 Definition of One-Sided Limits We write

$$\lim_{x \to a^-} f(x) = L$$

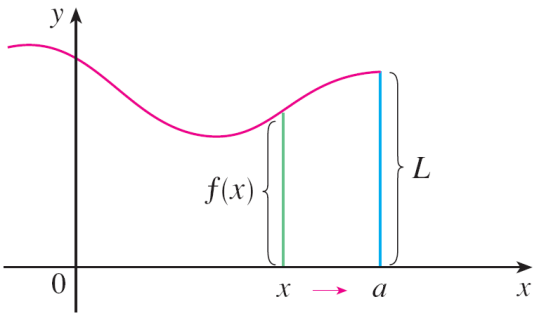
and say the **left-hand limit of $f(x)$ as x approaches a** [or the **limit of $f(x)$ as x approaches a from the left**] is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a with x *less than* a.

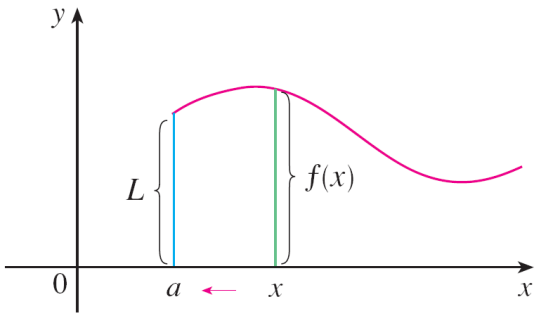
Notice that Definition 2 differs from Definition 1 only in that we require x to be less than a.

Similarly, if we require that x be greater than a, we get "the **right-hand limit of $f(x)$ as x approaches** a is equal to L" and we write

$$\lim_{x \to a^+} f(x) = L$$

Thus the notation $x \to a^+$ means that we consider only x *greater than* a. These definitions are illustrated below.





Example 2: Let $f(x) = \begin{cases} \sqrt{x} & if \ \ x \geq 0 \\ -2x + 1 & if \ \ x < 0 \end{cases}$.

(a.) Graph the function f.

(b.) Use the graph of f to find $\lim_{x \to 0^+} f(x)$ and $\lim_{x \to 0^-} f(x)$.

(c.) Find $\lim_{x \to 0} f(x)$, if it exist. If it does not exist, explain why?

3 $\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^-} f(x) = L$ and $\lim_{x \to a^+} f(x) = L$

Objective 3: Introduction to Infinite Limits.

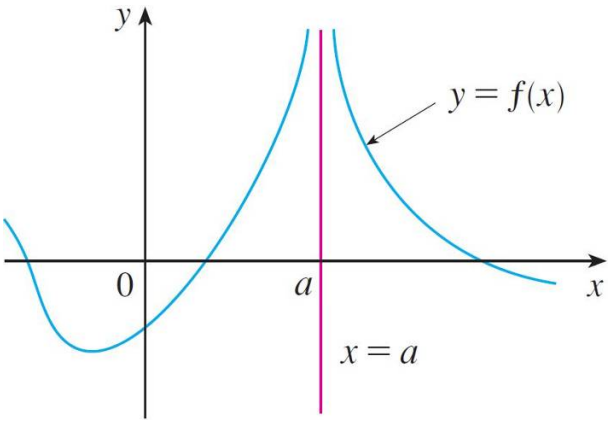
Example 3: Let $f(x) = \frac{1}{x^2}$. Find the following limits.

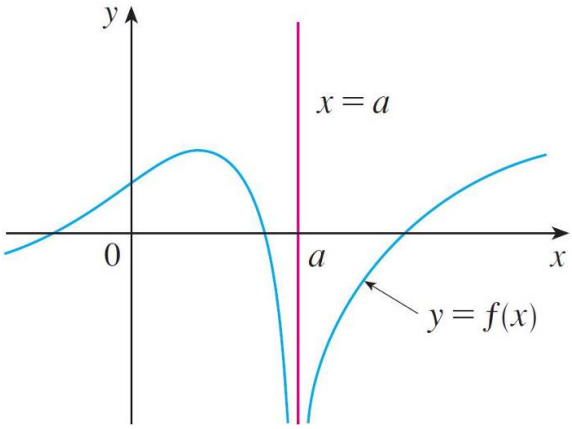
$(a.)\ \lim_{x \to 0^+} f(x)$ $(b.)\ \lim_{x \to 0^-} f(x)$ $(c.)\ \lim_{x \to 0} f(x)$

4 Intuitive Definition of an Infinite Limit Let f be a function defined on both sides of a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a, but not equal to a.





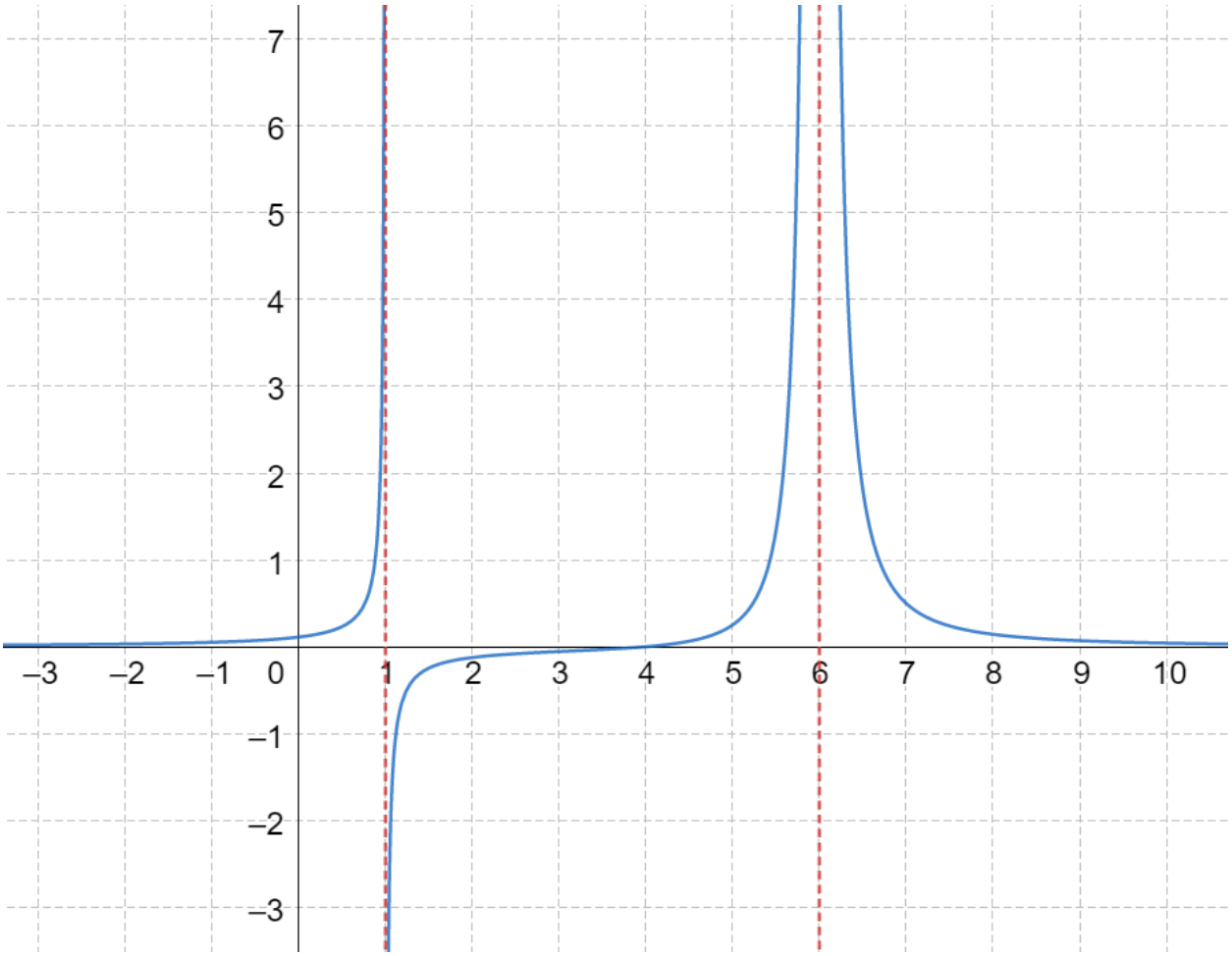
Objective 4: Using infinite limits to find vertical asymptotes.

6 Definition The vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

Example 4: Let $f(x) = \frac{x(x-4)}{x(x-1)(x-6)^2}$.



(a.) Find the vertical asymptotes of f.

(b.) Find $\lim_{x\to 1^+} f(x)$ and $\lim_{x\to 1^-} f(x)$.

(c.) Find $\lim_{x\to 6^+} f(x)$ and $\lim_{x\to 6^-} f(x)$.

Objective 5: Find limits analytically.

Example 5: Find the following limits.

$(a.)\ \lim_{x\to 0^+} \frac{1}{x}$

$\lim_{x\to 0^+} \frac{1}{x}$	
x	$\frac{1}{x}$

$(b.)\ \lim_{x\to 0^-} \frac{1}{x}$

$\lim_{x\to 0^-} \frac{1}{x}$	
x	$\frac{1}{x}$

$(c.)\ \lim_{x\to 2^+} \frac{1}{x-2}$

$(d.)\ \lim_{x\to 2^-} \frac{1}{x-2}$

Example 6: Find the following limits analytically.

$(a.)\ \lim_{x\to 2^+} \frac{3}{(x-2)^4}$

$(b.)\ \lim_{x\to 10^-} \frac{-8}{(x-10)^3}$

Example 7: Find the following limits analytically.

$(a.)\ \lim_{x\to 4^+} \dfrac{x+3}{x-4}$

$(b.)\ \lim_{x\to -8^+} \dfrac{5x+10}{x+8}$

$(c.)\ \lim_{x\to 1^-} \dfrac{2x}{(x-1)^2(x+3)}$

We can find the limits from Example 4 analytically. Let's see what that looks like.

Example 8: Find the following limits analytically.

$$(a.)\ \lim_{x\to 1^+} \frac{x(x-4)}{x(x-1)(x-6)^2}$$

Example 8: (continued)

$$(b.)\ \lim_{x\to 6^-} \frac{x(x-4)}{x(x-1)(x-6)^2}$$

Example 9: Find the following limit analytically.

$$\lim_{x \to \pi/2^-} \ln(\cos x)$$